Lecture 3:
Engineering or Hacking?

Prof. Martin Shepperd

Source: crAIyon

Notices

— For the labs this afternoon I have made a recording of me discussing a longer worked example based on covid using data from JHU.

— The video is in 2 parts on Loom here and here

— Bring headphones if possible.

Lecture Poll

Please use your mobile or computer to go to:

pollev.com/mshepperd

Agenda

1. Why does it matter? What does this code do?
2. Make it understandable and reproducible
3. Write code for humans
4. Manage the whole analysis
5. Week 3 goals

1. Why does it matter?

What does this code do?

```
g2 <- as.integer(readline(prompt="Enter g2: "))
if(g2<1){paste("Error")}else{g1<-1;
for(g3 in 1:g2) {g1 <- g1*g3}; paste("g1 is",g1)   }
```

Wisdom of the crowd

Improving the code

```
# Compute a factorial
n <- as.integer(readline(prompt =
  "Enter number for factorial computation:"))
if (n < 1) {
  paste("Error: enter number greater than zero")
} else {
  g1 <- 1

  for (g3 in 1:n) {
    g1 <- g1 * g3
  }
  paste("Factorial ",n," is", g1)
}
```

But is this necessary?

There's a built-in function, **`factorial()`** in Base R!

So we can reduce our code to ...

```
n <- as.integer(readline(
  prompt = "Enter number for factorial computation:"))
paste("Factorial ",n," is", factorial(n))
```

Data Analysis and the NHS



Background

— testing labs submit daily test reports to PHE as .csv files
— then merge data into Excel templates to upload to central system
— used .xls format (1986)
— has an upper limit ~65K rows
— each result uses multiple rows, so each template limited to ~1.4K cases
— once the limit is exceeded, further cases are simply left off

Consequence

Under-reported figures

From 25 Sept to 2 Oct

50,786

Cases initially reported by PHE

15,841

Unreported cases, missed due to IT error

---

8 days of incomplete data

1,980 cases per day, on average, were missed in that time

48 hours Ideal time limit for tracing contacts after positive test

Source: PHE and gov.uk



What lessons can we learn from this example?

Wisdom of the crowd

To errr is human ...

Our analysis of computational experiments[1][^2] (i.e., using machine learning for prediction):

— analysed 49 (101) scientific papers describing 2456 individual experimental results
— 22 / 49 (100/101) papers contain demonstrable **errors**
— include transcription, calculation and statistical errors
— not all errors are easy to detect

[1] Shepperd, M., et al. (2019). The Prevalence of Errors in Machine Learning Experiments. In International Conference on Intelligent Data Engineering and Automated Learning (pp102-109). Springer.
[^2]: DeStefanis, G., et al. (2026). An audit of machine learning experiments on software defect prediction, Empirical Software Engineering 31 (4).

2. Make it understandable and reproducible

Instead of imagining that our main task is to instruct a computer what to do, let us concentrate rather on explaining to human beings what we want the computer to do.

— Donald Knuth[4]

[4] Knuth, Donald E. 1984. "Literate Programming." The Computer Journal 27 (2): 97–111.

Understandability

It's important that the steps of an analysis can be understood e.g., our R Notebooks (or Jupyter, etc) should explain how and why we did things.

If data are to be reusable by others...

FAIR data principles[3]

— **F**indability
— **A**ccessibility
— **I**nteroperability
— **R**eusability

[3] Wilkinson, Mark D, et al. 2016. "The FAIR Guiding Principles for Scientific Data Management and Stewardship." Scientific Data 3 (1): 1–9.

A mystery data set

	ID	sex	score	date
1	A103	1	72	03/04/26
2	B217	2	58	07/04/26
3	C041	1	91	09/04/26
4	D305	2	66	12/04/26

What do 1 and 2 mean?
What is score measuring, and what are its units/range?
What does 03/04/26 mean — 3 April or 4 March?
What does ID identify?
What population do these observations come from?

Let's avoid mysteries (in data science)!

What is Meta-data?

- data about data
- meta-data describes how the data set is organised and the meanings of individual variables
- for example, we might say that the variable `PersonID`
  - comprises alpha-numeric characters (so it's a string)
  - to be valid it must be (i) unique and (ii) exactly 8 chars
- it facilitates meaningful processing and analysis

eproducibility and Replicability

— **Reproducibility**: data and code are available so others can repeat the analysis and obtain the same results.

— **Replicability**: repeating the study with new data produces similar findings.

— Minimally, we need reproducibility in data science.

3. Write code for humans

What has the greatest impact on R code 'literacy'?

Wisdom of the crowd

Readable code tips[6]

— names

— comments

— layout e.g., indentation and spacing

— prettyprinting

— user-defined functions

[6] See (Wickham, 2012) for many ideas about writing readable R code.

User-defined functions

— abstract (hide) details to help understandability e.g., `checkInput()`

— place related code in one place

— reuse

— maintainability i.e., only need to make changes in one place

User-defined function example

Define a function called halve

```
halve <- function(n) {
  # This function takes a number and divides it by two
  functionResult <- n / 2
  return(functionResult)
}
```

Use the function

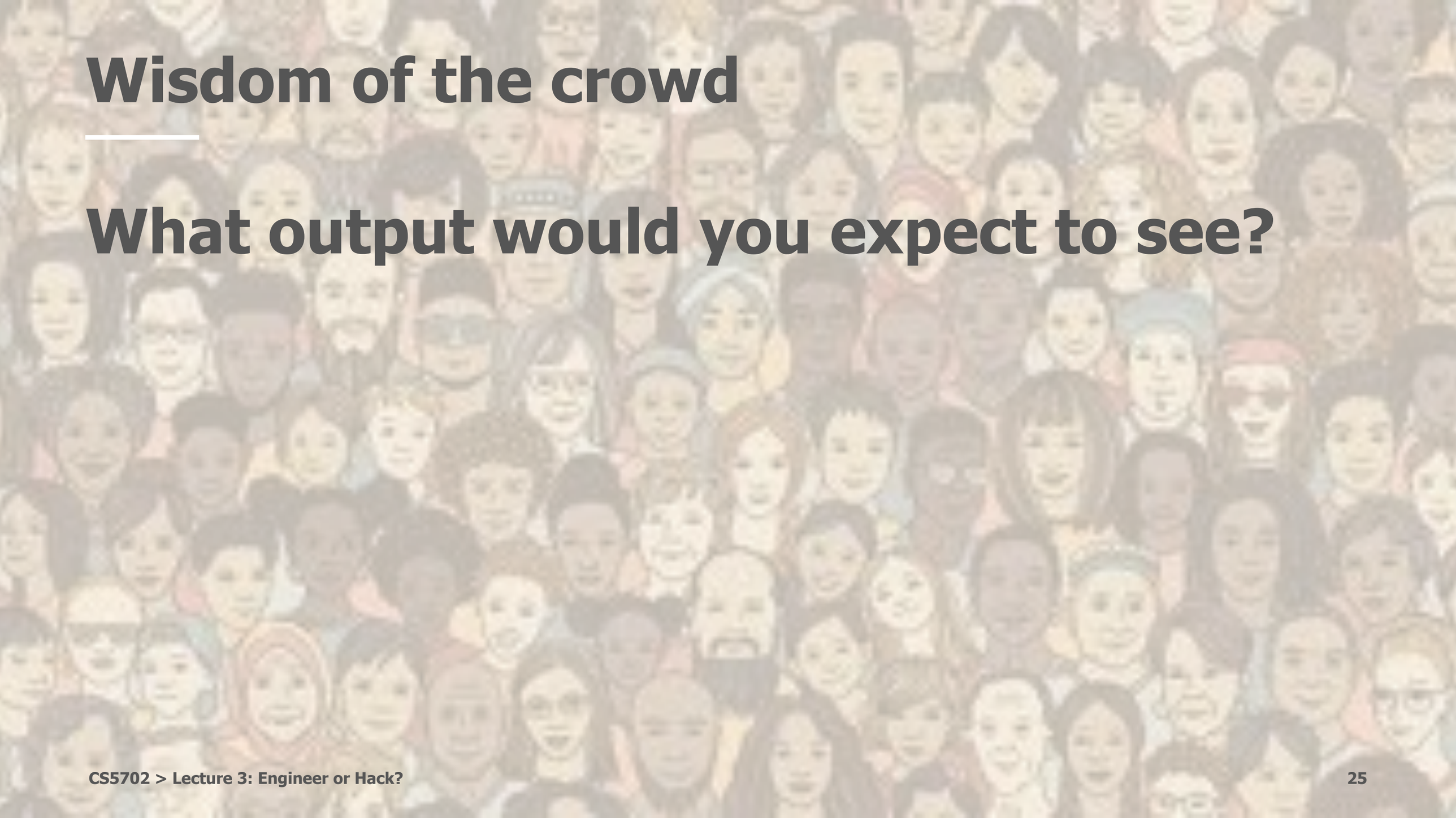
```
> halve(-8.3)
[1] -4.15
```

Parts of a function:

1. **Function Name**: how we call the function e.g., `head()`.
2. **Argument(s)**: when a function is called, we pass values to match the (0+) arguments e.g., `head(myDataFrame)`.
3. **Function Body**: contains a set of statements that defines what the function does. (For in-built functions these are hidden but for user defined-functions we need to provide them.)
4. **Return Value**: the result after being called and executing

Wisdom of the crowd

What output would you expect to see?



4. Manage the whole analysis

Why do we need to be disciplined?

— Multiple versions

— Dependencies

— Backup

— Wasted time

— Avoid careless mistokes



R packages and libraries

- R comprises **Base R** (what you get automatically)
- and can be extended with **packages** that can be downloaded
  - From cran.r-project.org/
  - or sometimes GitHub
  - ~20,000 packages on CRAN
  - contain vast number of extra functions

Installing packages[5]

1. You need to download a package **ONCE** to local storage

```
# Install tidyverse from CRAN to your computer
# NB Quotes needed for the package name
install.packages("tidyverse")
```

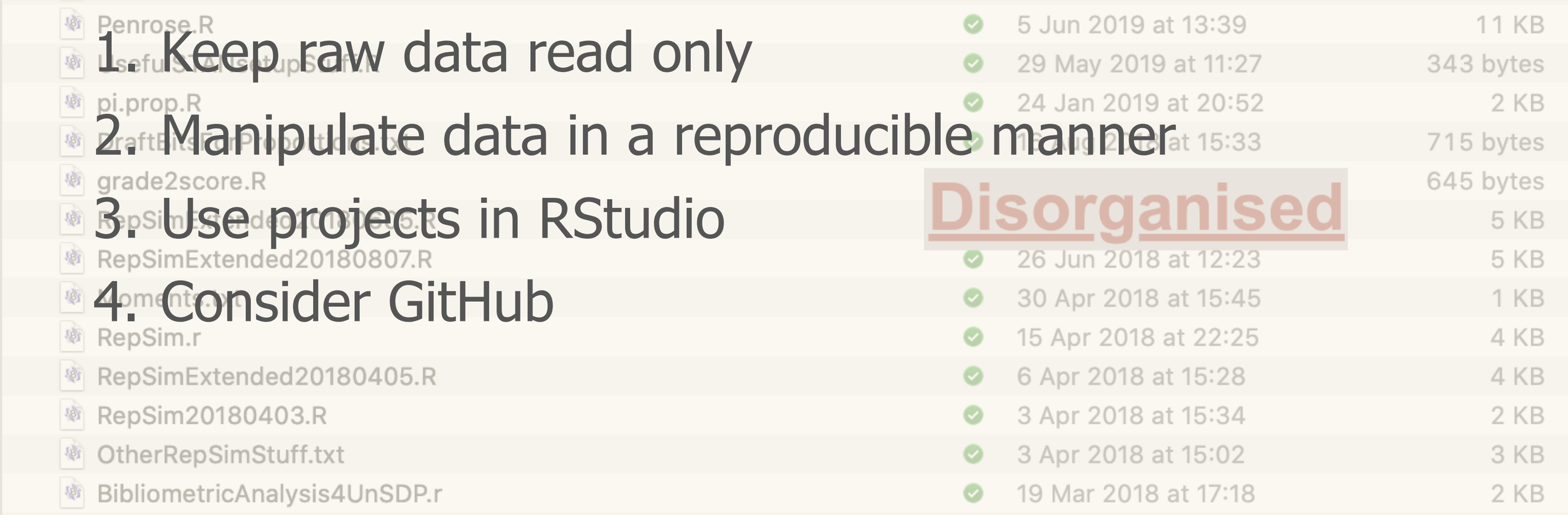
1. Load the functions in the package into your R session **EVERY** time you start R

```
# Load tidyverse from your library into RAM
# NB no quotes needed for the package name
library(tidyverse)
```

[5] For thorough coverage on packages see Thomas Neitmann's blog at https://thomasadventure.blog/posts/install-r-packages/.

Final tips

1. Keep raw data read only
2. Manipulate data in a reproducible manner
3. Use projects in RStudio
4. Consider GitHub



5. Week 3 goals

By the end of this week you should:
[ ] have a thorough appreciation of the need for engineered data analysis
[ ] aware of different tactics to improve the understandability of quality of your data analysis
[ ] be able to implement simple user-defined R functions

Further Reading

1. For more details on R functions see Modern Data, chapter three
2. Alternatively see chapter 5 from Kabacoff (2015) R in Action: Data Analysis and Graphics with R
3. For thorough coverage on packages see Thomas Neitmann's blog at https://thomasadventure.blog/posts/install-r-packages/.
4. For some thoughts on classic errors ("bad smells") in using data analytics in scientific research: Menzies, T., & Shepperd, M. (2019). “Bad smells” in software analytics papers. Information and Software Technology, 112, 35-47.
5. For detailed thoughts on R style and readability see: Wickham, H. (2012). "Style Guide." http://adv-r.had.co.nz/Style.html.

Questions